Key Boundary & Control Classes

Boundary Classes

Class	Responsibility	Control
SearchHomeScreenUI	Gets destination, arrival time and duration. Shows the top 5 results as a list or map	RecommendationController
CarparkDetailViewUI	Shows the carpark record, rates and predicted lots. Has navigate, favourite and report actions.	RecommendationController,Navgation Controller
SettingsUI	Shows preferences and saved locations	PreferencesController
SessionUI	Shows the active session and session history	SessionController
FavouritesUI	Shows favourite carparks	FavouritesController
ReportSubmissionModelUI	Gets the report category and description	ReportController
OperatorLoginUI	Operator log in form in the Operator tab	AuthenticationController
OperatorConsoleUI	Shows the report queue, overrides and data source health	OperatorAdminController, ReportController
CarparkDataInterface	Connects to LTA DataMall, data.gov.sg and URA	ExternalDataController
MapServiceInterface	Connects to Google Maps and OneMap	RecommendationController, PreferencesController
NavigationIntentInterface	Sends the carpark location to the device navigation app	NavigationController
NotificationInterface	Schedules and cancels the parking reminder	SessionController

Control Classes

Class	Responsibility	Use Case
RecommendationControll er	Finds nearby carparks. Excludes closed carparks and carparks below vehicle height. Predicts lots at arrival. Estimates cost. Scores and ranks the carparks. Reranks the results. Gets Carpark Details	UC-01, UC-02
NavigationController	Starts navigation. Shows the address to copy when no navigation app is installed.	UC-03
ReportController	Submits anonymous reports. Applies the daily limit with the installation token. Rejects reports for carparks taht are not in the dataset. Records the operator's action on each report.	UC-05, UC-06
AuthenticationController	Logs the operator in. Issues the session token. Revokes the token on log out or timeout.	UC-05
OperatorAdminController	Applies and withdraws overrides. Records each override with the reason and time. Protected overridden records from the daily refresh.	UC-06, UC-07
APIHealthMonitor	Collects per-source error counts, sync times and cache age. Highlights stale sources.	UC-08
ExternalDataController	Polls availability every 60 seconds. Refreshes reference data every day. Runs the manual refresh. Reads source health and refresh attempts.	UC-09
PreferencesController	Validates and saves preferences and saved locations on the device. Erases all local data.	FR1.1 - FR1.10
SessionController	Starts, extends and ends the session. Calculates cost. Schedules the reminder.	FR6.1 - FR6.11
FavouritesController	Adds and removes favourites on the device.	FR7.1 - FR7.5

Stereotype	Class Name	Responsibilities	Collaborators
Boundary	SearchHomeScreenUI	Captures destination string/pin and parking duration inputs; displays ranked candidate cards.	RecommendationController
Boundary	CarparkDetailViewUI	Renders rate tables, lot availability timestamp, and navigation trigger	RecommendationController, NavigationController
Boundary	OperatorConsoleUI	Renders operator login, active error report queue, override inputs, and API health panels.	OperatorAdminController, AuthenticationController
Boundary	ReportSubmissionModalUI	Collects anonymous user data error reports and submission categories.	ReportController
Boundary	LTADataMallAdapter	Ingests 60s external availability polling feeds.	ExternalDataController
Boundary	NavigationIntentLauncher	Executes OS-level app intent to open external navigation.	NavigationController
Control	AuthenticationController	Verifies operator login credentials over TLS; issues session tokens.	OperatorAccount, OperatorConsoleUI
Control	OperatorAdminController	Manages manual overrides, protects overridden records from daily refreshes, and reads API telemetry.	OperatorOverride, Carpark, APIHealthMonitor
Control	ReportController	Enforces the 5-report daily submission quota and queues reports for operator review.	ErrorReport, OperatorAdminController
Control	RecommendationControl	Filters out	Carpark,

	ler	closed/overridden carparks and computes Best Overall recommendations	ScoringEngine, CostEngine
Control	APIHealthMonitor	Collects per-source error counters, sync timestamps, and cache staleness metrics	ExternalDataControl ler, OperatorConsoleUI
Entity	Carpark, OperatorAccount, OperatorOverride, ErrorReport, AvailabilityRecord, RateSchedule	Domain entities maintaining persistent properties and relationships.	System Controllers